

Adding & Subtracting Fractions

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS When two fractions have the **same (common) denominator**, they are cut into the same size pieces. To **add** or **subtract** them, add or subtract the **numerators** (the top numbers) and **keep the denominator the same** (the bottom number does not change). Example: $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$. Example: $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$.

Round 1 — Retrieve It

1. What does the denominator (bottom number) tell you about the pieces?

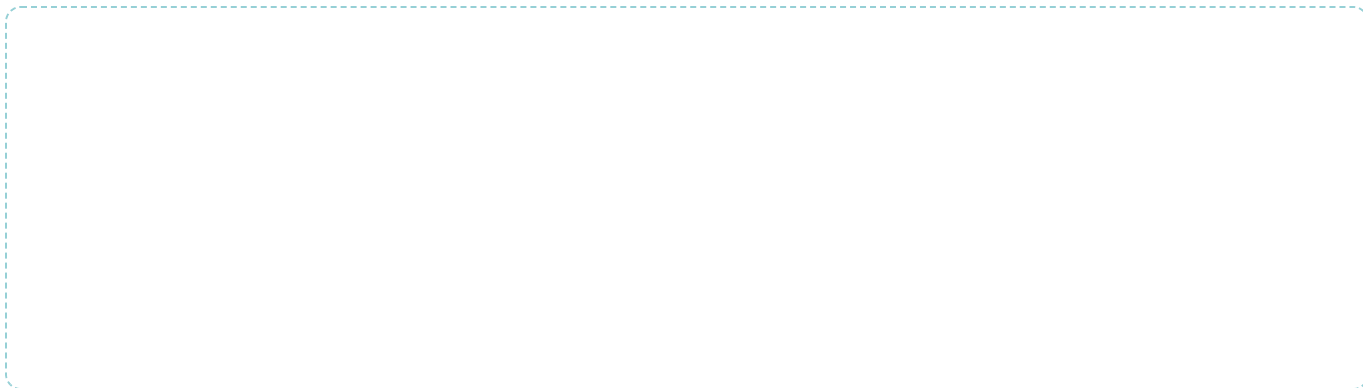
2. When denominators are the same, what do you do with the numerators to add?

3. Does the denominator change when you add fractions with the same denominator?

4. What is $\frac{1}{4} + \frac{2}{4}$?

Round 2 — Sketch It

Draw a bar split into 4 equal parts. Shade $\frac{1}{4}$ one way and $\frac{2}{4}$ another way to show $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$.



Adding & Subtracting Fractions — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. Solve $\frac{1}{4} + \frac{2}{4}$. Show the numerators you added and the denominator you kept.

2. Solve $\frac{3}{5} - \frac{1}{5}$. Show your work.

3. Solve $\frac{2}{6} + \frac{3}{6}$. What denominator did you keep?

Round 4 — Reason from Evidence

A student solved $\frac{1}{4} + \frac{2}{4}$ and wrote $\frac{3}{8}$ by adding the tops and the bottoms. Explain the mistake and give the correct answer.

Adding & Subtracting Fractions — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain why the denominator stays the same when you add fractions with a common denominator. Use the word pieces.

C**Connect**

Connect this to sharing a pizza cut into equal slices. If you eat 3 of 8 slices and a friend eats 2 of 8, how much is gone?

E**Extend**

Solve $5/8 - 2/8$, then write a subtraction story that matches your problem.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Adding & Subtracting Fractions

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. It tells how many equal pieces the whole is cut into (the size of each piece).
2. Add the numerators (the top numbers).
3. No, the denominator stays the same.
4. $\frac{3}{4}$.

Sketch It — look for

A bar in 4 equal parts with 3 parts shaded in total (1 shaded one way, 2 shaded another), showing $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$.

Apply It

1. $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$ (add $1 + 2 = 3$ for the numerator; keep 4).
2. $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$ (subtract $3 - 1 = 2$; keep 5).
3. $\frac{2}{6} + \frac{3}{6} = \frac{5}{6}$ (kept denominator 6).

Reason from Evidence — look for

You do not add the denominators; the pieces are still fourths. Only the numerators are added, so $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$, not $\frac{3}{8}$.

ACE — Extend look-for

$\frac{5}{8} - \frac{2}{8} = \frac{3}{8}$, with a matching subtraction story (for example, having $\frac{5}{8}$ of something and giving away or using $\frac{2}{8}$).

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.